Applied Statistics end-of-module assignment
MSc in Statistics, Imperial College London

I, 06060027, certify that this assessed coursework is my own work, unless otherwise acknowledged, and includes no plagiarism. I have not discussed my coursework with anyone else except when seeking clarification with the module lecturer via email or on MS Teams. I have not shared any code underlying my coursework with anyone else prior to submission.

06060027

Contents

1 Question 1 - independence of participants

In practice, several forms of clustering could violate the independence assumption.

(1) Family / household clustering.
Members of the same household share genetic and environmental factors (e.g., passive smoking, indoor air quality), so cancer outcomes and lung capacities may be positively correlated within households even after conditioning on (A_i, W_i, η_i). This intra-cluster correlation reduces the effective sample size and leads to underestimation of standard errors and inflated Type I error if independence is assumed.

(2) Workplace / neighbourhood clustering.
Participants working or living in the same area share unmeasured exposures ($PM_{2.5}$, NO_2, local smoking culture). This can induce residual correlation in C_i and L_i across individuals in the same spatial cluster and create spatial confounding (both smoking intensity and cancer risk higher in certain areas). Ignoring this dependence may produce biased effect estimates and overly optimistic uncertainty intervals.

(3) Measurement batch effects.
Lung capacity may be measured in batches (same spirometer, same technician), inducing batch-specific biases or calibration errors. Then

$$L_i^{\text{obs}} = L_i^{\text{true}} + \Delta_{k(i)} + \epsilon_i$$

with a shared batch term $\Delta_{k(i)}$, so measurement errors are correlated within batch. This violates the assumption $\mathbb{E}[\epsilon_i \epsilon_j] = 0$ for $i \neq j$, leading again to underestimated standard errors unless batch effects or cluster-robust methods are used.

2 Question 2 - modelling smoking status vs intensity

2.1 (a) Information loss from binarization

Replacing the continuous intensity η_i with the binary status $S_i = ⊮(\eta_i > 0)$ induces a clear loss of statistical information. The continuous η_i carries variation across $[0, 26]$, so any regression or likelihood using η_i exploits this spread (larger Fisher information and ability to estimate a dose,response slope or curvature).

By contrast, S_i collapses all positive values to a single category, so the analysis becomes a simple "smoker vs non,smoker" comparison. This discards within,smoker variability, reduces the variance of the covariate, and therefore lowers the information about the effect of smoking, leading to less precise and less sensitive inference on the association with cancer.

2.2 (b) Two likelihoods and sufficiency

Full likelihood: Assuming participants are independent, each C_i follows a Bernoulli distribution with parameter p if $S_i = 1$ or q if $S_i = 0$. The joint likelihood factorizes as

$$P(C_{1:N}|p, q, S_{1:N}) = \prod_{i=1}^{N} [p^{S_i}(1-p)^{S_i}]^{C_i}[q^{1-S_i}(1-q)^{1-S_i}]^{C_i} \times [p^{S_i}(1-p)^{S_i}]^{1-C_i}[q^{1-S_i}(1-q)^{1-S_i}]^{1-C_i}$$

which simplifies by grouping smokers and non-smokers:

$$P(C_{1:N}|p, q, S_{1:N}) = p^{N_{s,c}}(1-p)^{N_s - N_{s,c}} \times q^{N_{\neg s,c}}(1-q)^{N_{\neg s} - N_{\neg s,c}}$$

where $N_s = \sum_i S_i = 10$ is the total number of smokers, $N_{\neg s} = 36 - 10 = 26$ is the number of non-smokers, $N_{s,c} = \sum_i S_i C_i = 6$ is the number of smokers with cancer, and $N_{\neg s,c} = \sum_i (1 - S_i) C_i = 2$ is the number of non-smokers with cancer.

Grouped likelihood: When working with the contingency table $(N_{s,c}, N_{\neg s,c})$ given the marginals $(N_s, N_{\neg s})$, we recognize that within each smoking group, the number of cancers follows a binomial distribution. There are $\binom{N_s}{N_{s,c}}$ ways to assign $N_{s,c}$ cancers among N_s smokers, each with probability $p^{N_{s,c}}(1-p)^{N_s - N_{s,c}}$, and similarly for non-smokers. Thus

$$P(N_{s,c}, N_{\neg s,c}|p, q, N_s, N_{\neg s}) = \binom{N_s}{N_{s,c}} p^{N_{s,c}}(1-p)^{N_s - N_{s,c}} \times \binom{N_{\neg s}}{N_{\neg s,c}} q^{N_{\neg s,c}}(1-q)^{N_{\neg s} - N_{\neg s,c}}$$

The grouped likelihood differs from the full likelihood only by the multiplicity term

$$h(\text{data}) = \binom{N_s}{N_{s,c}}\binom{N_{\neg s}}{N_{\neg s,c}},$$

which counts the number of distinct individual-level cancer labelings consistent with the same 2×2 contingency table. Hence the individual-level likelihood and the count likelihood are **proportional as functions of** (p, q):

$$P(C_{1:N} \mid p, q, S_{1:N}) \;\propto\; P(N_{s,c}, N_{\neg s,c} \mid p, q, N_s, N_{\neg s}),$$

with proportionality factor free of (p, q). Therefore they generate identical likelihood ratios, posteriors and Bayes factors, because all inferential updates depend on the data only through the likelihood up to a (p, q)-free factor. Equivalently, the joint pmf admits the Fisher–Neyman factorization

$$f(\text{data} \mid p, q) = h(\text{data})\, g(T(\text{data}); p, q),$$

where $T = (N_{s,c}, N_{\neg s,c}, N_s, N_{\neg s})$ and

$$g(T; p, q) = p^{N_{s,c}}(1-p)^{N_s - N_{s,c}} q^{N_{\neg s,c}}(1-q)^{N_{\neg s} - N_{\neg s,c}}.$$

Thus T is sufficient (indeed minimal, up to one-to-one reparameterisation) for (p, q).

2.3 (c) Posterior for p and q: Beta-Binomial conjugacy

2.3.1 Prior and posterior

We place independent uniform priors on the two cancer probabilities

$$p \sim \text{Beta}(1, 1), \qquad q \sim \text{Beta}(1, 1),$$

so $\pi(p, q) = \pi(p)\pi(q)$ and each prior is weakly informative and proper on $[0, 1]$.

For smokers $(N_s = 10, N_{s,c} = 6)$ the likelihood is $P(N_{s,c} \mid p) \propto p^{N_{s,c}}(1-p)^{N_s - N_{s,c}}$, so

$$\pi(p \mid \text{data}) \propto p^{1-1}(1-p)^{1-1} \times p^{N_{s,c}}(1-p)^{N_s - N_{s,c}} = p^{N_{s,c}}(1-p)^{N_s - N_{s,c}},$$

i.e.

$$p \mid \text{data} \sim \text{Beta}(\alpha_p, \beta_p), \qquad \alpha_p = N_{s,c} + 1,\; \beta_p = N_s - N_{s,c} + 1 = \text{Beta}(7, 5).$$

Analogously, for non-smokers $(N_{\neg s}, N_{\neg s,c})$ we obtain

$$q \mid \text{data} \sim \text{Beta}(\alpha_q, \beta_q), \qquad \alpha_q = N_{\neg s,c} + 1,\; \beta_q = N_{\neg s} - N_{\neg s,c} + 1 = \text{Beta}(3, 25).$$

group	N	cancer	no_cancer
Smokers	10	6	4
Non−smokers	26	2	24

parameter	mean	median	lower95	upper95
p = P(C|S)	0.583	0.588	0.308	0.833
q = P(C|not S)	0.107	0.098	0.024	0.243

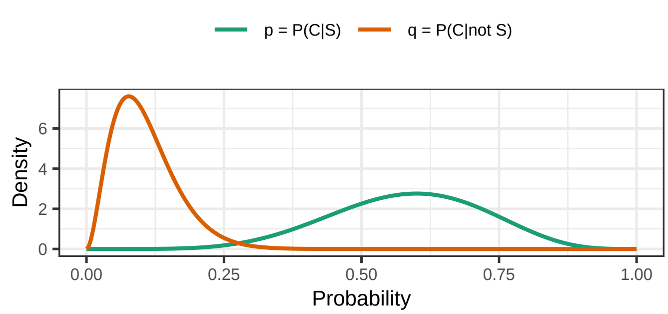

Figure 1: Left: Counts and posterior summaries (stacked). Right: Marginal posterior densities for p and q.

The posterior for p (smokers) is centred around 0.58 with moderate spread, while the posterior for q (non-smokers) is centred much lower, around 0.11. The two Beta densities overlap only slightly: most of the mass of p lies above most of the mass of q, providing strong visual evidence that $p > q$.

Formally, since the posteriors factor as $\pi(p, q \mid \text{data}) = \pi(p \mid \text{data})\, \pi(q \mid \text{data})$, we can compute

$$P(p > q \mid \text{data}) = \int_0^1 \int_0^p \text{Beta}(p; 7, 5)\, \text{Beta}(q; 3, 25)\, dq\, dp,$$

or equivalently

$$P(p > q \mid \text{data}) = \int_0^1 \text{Beta}(p; 7, 5)\, F_{\text{Beta}}(p; 3, 25)\, dp,$$

which numerically gives $P(p > q \mid \text{data}) \approx 0.999$ (posterior odds of about 1353.6:1 for $p > q$). Thus the data provide clear Bayesian evidence that the cancer risk is higher for smokers than for non-smokers.

2.4 (d) Model comparison via Bayes factors

We compare three models for the relationship between smoking (S) and lung cancer (C):

1. M_1**:** $p > q$ (harmful effect): smoking increases cancer risk.
2. M_2**:** $p = q$ (null effect): smoking has no association with cancer.
3. M_3**:** $p < q$ (protective effect): smoking decreases cancer risk.

Working with the summary statistics $N_{s,c}, N_{\neg s,c}, N_s, N_{\neg s}$, the likelihood for (p, q) (under any model) is

$$P(N_{s,c}, N_{\neg s,c} \mid p, q, N_s, N_{\neg s}) = \binom{N_s}{N_{s,c}} p^{N_{s,c}} (1-p)^{N_s - N_{s,c}} \binom{N_{\neg s}}{N_{\neg s,c}} q^{N_{\neg s,c}} (1-q)^{N_{\neg s} - N_{\neg s,c}}.$$

Let

$$K = \binom{N_s}{N_{s,c}} \binom{N_{\neg s}}{N_{\neg s,c}}$$

denote the binomial factor. It does not depend on p or q, but I keep it explicitly so that the marginal likelihoods are for the *summary counts*, as requested in the question.

The marginal likelihood under model M_k is

$$P(\text{data} \mid M_k) = \int_{\Theta_k} P(N_{s,c}, N_{\neg s,c} \mid p, q, N_s, N_{\neg s})\, \pi(p, q \mid M_k)\, dp\, dq,$$

where Θ_k and $\pi(\cdot \mid M_k)$ are the parameter space and prior for M_k.

2.4.1 Model M_1 $(p > q)$

The prior density is uniform over the triangle $\{(p, q) : 0 < q < p < 1\}$. Since this region has area $1/2$, the normalised prior density is

$$\pi(p, q \mid M_1) = 2 \quad \text{for } 0 < q < p < 1.$$

Thus

$$P(\text{data} \mid M_1) = K \int_0^1 \int_0^p 2\, p^{N_{s,c}} (1-p)^{N_s - N_{s,c}} q^{N_{\neg s,c}} (1-q)^{N_{\neg s} - N_{\neg s,c}}\, dq\, dp,$$

which I evaluate numerically in R.

2.4.2 Model M_2 $(p = q)$

Under M_2 we set $p = q = \theta$ and place a uniform prior on $\theta \in [0, 1]$. Pooling smokers and non-smokers, we observe $N_c = N_{s,c} + N_{\neg s,c}$ cancers out of $N = N_s + N_{\neg s}$ individuals, so

$$P(\text{data} \mid M_2) = K \int_0^1 \theta^{N_c} (1-\theta)^{N - N_c}\, d\theta = K\, B(N_c + 1, N - N_c + 1) = K\, \frac{N_c!\, (N - N_c)!}{(N+1)!}.$$

For our data ($N_c = 8$, $N = 36$),

$$P(\text{data} \mid M_2) = K\, \frac{8!\, 28!}{37!}.$$

2.4.3 Model M_3 ($p < q$)

By symmetry of the prior, the density under M_3 is uniform over the triangle $\{(p, q) : 0 < p < q < 1\}$:

$$\pi(p, q \mid M_3) = 2 \quad \text{for } 0 < p < q < 1.$$

The likelihood is the same as above, but the integration region is now $p < q$:

$$P(\text{data} \mid M_3) = K \int_0^1 \int_p^1 2\, p^{N_{s,c}} (1-p)^{N_s - N_{s,c}} q^{N_{\neg s,c}} (1-q)^{N_{\neg s} - N_{\neg s,c}} \, dq\, dp,$$

which I again evaluate numerically in R.

Model	Marginal Likelihood
M1: p > q	6.73e-03
M2: p = q	6.10e-05
M3: p < q	4.97e-06

Table 1: Marginal likelihoods (scientific notation)

Model	M1: p > q	M2: p = q	M3: p < q
M1: p > q	1.00	110.39	1353.59
M2: p = q	0.01	1.00	12.26
M3: p < q	0.00	0.08	1.00

Table 2: Pairwise Bayes factors (rows = numerator, columns = denominator)

To quantify evidence for a smoking,cancer link we compute the Bayes factor. The Bayes factor $B_{1,2} = P(\text{data} \mid M_1)/P(\text{data} \mid M_2) \approx 110.4$ means the data are about 110 times more likely under the harmful model (M_1) than under the null (M_2). By conventional thresholds this constitutes moderate evidence in favour of M_1 given the small sample. Similarly, $B_{1,3} \approx 1354$ strongly favours the harmful model M_1 over the protective model M_3.

3 Question 3 - calibration noise model

3.0.1 Part (i): Normalizability

A normalising constant exists iff

$$Z(\beta, \sigma) = \int_{-\infty}^{\infty} \left[1 + \frac{(L - L^* - \Delta)^2}{\sigma^2}\right]^{-\beta} dL < \infty.$$

Let $u = (L - L^* - \Delta)/\sigma$, so $dL = \sigma\, du$ and

$$Z(\beta, \sigma) = \sigma \int_{-\infty}^{\infty} (1 + u^2)^{-\beta}\, du.$$

The integrand is bounded on $|u| \le 1$, so only tails matter. For $|u| > 1$, $(1 + u^2)^{-\beta} \le u^{-2\beta}$, hence

$$\int_{|u|>1} (1 + u^2)^{-\beta} du \le 2 \int_1^{\infty} u^{-2\beta} du < \infty \iff \beta > 1/2.$$

Thus the density is proper iff $\beta > 1/2$.

3.0.2 Part (ii): Defined mean

If $\beta > 1/2$, symmetry implies (when it exists) $\mathbb{E}[L] = L^* + \Delta$. Existence of the first absolute moment requires

$$\mathbb{E}[|L - L^* - \Delta|] \propto \int_{-\infty}^{\infty} |u|(1 + u^2)^{-\beta}\, du < \infty.$$

For $|u| > 1$, $|u|(1 + u^2)^{-\beta} \le |u|^{1-2\beta}$, so

$$\int_1^{\infty} u^{1-2\beta} du < \infty \iff \beta > 1.$$

Hence the mean exists iff $\beta > 1$.

3.0.3 Part (iii): Defined variance

If $\beta > 1$, the variance exists iff

$$\mathbb{E}[(L - L^* - \Delta)^2] \propto \int_{-\infty}^{\infty} u^2(1+u^2)^{-\beta}\, du < \infty.$$

For $|u| > 1$, $u^2(1+u^2)^{-\beta} \leq u^{2-2\beta}$, so

$$\int_1^{\infty} u^{2-2\beta} du < \infty \iff \beta > 3/2.$$

Therefore $\mathrm{Var}(L)$ is finite iff $\beta > 3/2$.

3.0.4 Shortcut with Student-t distribution

Actually, we could have worked with an interesting transformation to recognize our distribution as a Student-t family, making analysis easier. Let $u = (L - L^* - \Delta)/\sigma$, so the density in u becomes:

$$p(u) = \frac{1}{Z(\beta)}\,(1+u^2)^{-\beta}$$

The standard Student-t_ν density has the form: $p_t(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2}$

Setting $x = \sqrt{\nu}\,u$ gives $1 + x^2/\nu = 1 + u^2$, so: $(1+u^2)^{-\beta} = \left(1+\frac{x^2}{\nu}\right)^{-\beta}$

Comparing with the Student-t kernel $\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2}$, we identify: $\beta = \frac{\nu+1}{2} \quad \iff \quad \nu = 2\beta - 1.$

Thus in u-space our distribution is Student-$t_{2\beta-1}$, and in L-space it is shifted-scaled: $L = L^* + \Delta + \sigma\, T, \qquad T \sim t_{2\beta-1}$.

For Student-t_ν, moment existence is well-characterized: Mean exists iff $\nu > 1 \iff \beta > 1$; Variance exists iff $\nu > 2 \iff \beta > 3/2$; The k-th moment exists iff $\nu > k \iff \beta > (k+1)/2$

This confirms our earlier findings: normalizability requires $\beta > 1/2$ (corresponding to $\nu > 0$, an improper distribution for $\nu \leq 0$); a defined mean requires $\beta > 1$; defined variance requires $\beta > 3/2$.

3.0.5 Which criteria are necessary for a valid noise model?

Normalizability ($\beta > 1/2$) is essential: otherwise $\tilde{P}$ cannot be normalised (the integral diverges), so the likelihood/posterior are improper and Bayesian inference is not well-defined.

Finite mean ($\beta > 1$) is not required for a valid likelihood, but it changes interpretation: for $\beta \in (1/2, 1]$ the model is proper, yet Δ is no longer an *expected* bias and moments like $\mathbb{E}[\Delta \mid \text{data}]$ may not exist (median/mode-based inference still does).

Finite variance ($\beta > 3/2$) is not required but impacts uncertainty and asymptotics: for $\beta \in (1, 3/2]$ the distribution is proper with a defined centre, but has infinite variance, yielding heavy-tailed posteriors/intervals and potentially undermining large-sample normal approximations (often tied to finite Fisher information).

3.1 (b) Negative measurement probabilities

3.1.1 Part (i): Problem in principle

The support of the distribution is the entire real line: $L \in (-\infty, \infty)$. Since lung capacity is a physical volume measured in litres, negative values are **physically impossible** and represent a violation of the domain constraint $L > 0$ resulting in a model misspecification issue.

3.1.2 Part (ii): Importance in practice

The calibration sample has $M = 40$ measurements with $\min(L_{1:M}) = 2.48$ L, $\max(L_{1:M}) = 4.6$ L, $\bar{L} = 3.73$ L, $s_L = 0.4$ L. Given $L^* = 3.44$ L, the empirical bias is $\bar{L} - L^* \approx 0.29$ L, so the data are centred near 3.5 L and never close to 0.

Under the Student,t representation with $\nu = 2\beta - 1$,

$$T = \frac{L - (L^* + \Delta)}{\sigma} \sim t_\nu,$$

the impossible region $L \leq 0$ corresponds to $P(L \leq 0) = P\Big(T \leq \frac{0 - L^* - \Delta}{\sigma}\Big)$. For parameter values consistent with the data, e.g. - $\Delta = 0$, $\sigma = 0.4$ L, $\beta = 3$ ($\nu = 5$): $T_0 \approx -8.6$, $P(T \leq T_0) \approx 2 \times 10^{-6}$; - $\Delta = 0.2$, $\sigma = 0.6$ L, $\beta = 2$ ($\nu = 3$): $T_0 \approx -6.1$, $P(T \leq T_0) \approx 5 \times 10^{-5}$,

the event $L \leq 0$ lies about 6,9 "standard deviations" into the left tail, with probability always $< 10^{-4}$.

For comparison, the smallest observed value $\min(L_{1:M}) = 2.48$ L corresponds under, say, Case 2 to

$$T_{\min} = \frac{2.48 - 3.54}{0.5} \approx -2.12,$$

only about two standard deviations below the centre, typical for $M = 20$.

Overall, across realistic (Δ, σ, β) compatible with the data, $P(L < 0)$ is negligible in practice. The model's support on $(-\infty, \infty)$ is a purely theoretical misspecification with no practical impact on inference for (Δ, σ, β) in the region where the posterior concentrates.

3.2 (c) Posterior propriety under prior 1

With the improper prior $\tilde{\pi}_1(\Delta, \sigma, \beta) = \Theta(\sigma)\,\Theta(\beta - 2)$, the unnormalised posterior is

$$\tilde{p}(\Delta, \sigma, \beta \mid L_{1:M}) \propto \Theta(\sigma)\,\Theta(\beta - 2) \left[\frac{\Gamma(\beta)}{\sigma\sqrt{\pi}\,\Gamma(\beta - 1/2)}\right]^M \prod_{m=1}^{M} \left[1 + \frac{(L_m - L^* - \Delta)^2}{\sigma^2}\right]^{-\beta},$$

with $\sigma > 0,\ \beta \geq 2$. Assume $\mathrm{Var}(L_{1:M}) > 0$.

Let

$$g(\Delta \mid \sigma, \beta) = \prod_{m=1}^{M} \left[1 + \frac{(L_m - L^* - \Delta)^2}{\sigma^2}\right]^{-\beta}.$$

3.2.1 Step 1: Integrate over Δ

Write $a_m = L_m - L^*$. On bounded Δ-intervals, g is continuous and bounded, hence integrable. For tails, if $|\Delta| > 2|a_m|$ then $|a_m - \Delta| \geq |\Delta|/2$, so with $R = \max_m 2|a_m|$,

$$g(\Delta \mid \sigma, \beta) \leq \left[1 + \frac{\Delta^2}{4\sigma^2}\right]^{-\beta M} \qquad (|\Delta| > R).$$

With $u = \Delta/(2\sigma)$,

$$\int_{|\Delta| > R} g(\Delta \mid \sigma, \beta)\, d\Delta \leq 2\sigma \int_{\mathbb{R}} (1 + u^2)^{-\beta M}\, du < \infty,$$

since $\beta M > 1/2$ (here $\beta \geq 2$). Hence

$$J(\sigma, \beta) := \int_{\mathbb{R}} g(\Delta \mid \sigma, \beta)\, d\Delta < \infty \qquad (\sigma > 0,\ \beta \geq 2).$$

After integrating out Δ, the (σ, β)-kernel is

$$k(\sigma, \beta) \propto \Theta(\sigma)\,\Theta(\beta - 2) \left[\frac{\Gamma(\beta)}{\sigma\sqrt{\pi}\,\Gamma(\beta - 1/2)}\right]^M J(\sigma, \beta).$$

3.2.2 Step 2: Joint tail asymptotics and divergence

Using $u = \Delta/\sigma$,

$$J(\sigma, \beta) = \sigma \int_{\mathbb{R}} \prod_{m=1}^{M} \left[1 + \left(u - \frac{a_m}{\sigma}\right)^2\right]^{-\beta} du \sim \sigma\, C(\beta, M) \quad (\sigma \to \infty),$$

where

$$C(\beta, M) = \int_{\mathbb{R}} (1 + u^2)^{-\beta M} du = \sqrt{\pi}\,\frac{\Gamma(\beta M - 1/2)}{\Gamma(\beta M)} \sim \text{const} \cdot \beta^{-1/2} \quad (\beta \to \infty).$$

Also $\Gamma(\beta)/\Gamma(\beta - 1/2) \sim \text{const} \cdot \beta^{1/2}$, so

$$\left[\frac{\Gamma(\beta)}{\sigma\sqrt{\pi}\,\Gamma(\beta - 1/2)}\right]^M \sim \text{const} \cdot \sigma^{-M} \beta^{M/2}.$$

Therefore in the joint tail $\sigma, \beta \to \infty$,

$$k(\sigma, \beta) \sim \text{const} \cdot \sigma^{1-M} \beta^{(M-1)/2}.$$

Consider $\mathcal{R} = \{(\sigma, \beta) : \beta > B,\ c_1\sqrt{\beta} \le \sigma \le c_2\sqrt{\beta}\}$. On $\mathcal{R}$, $\sigma \asymp \sqrt{\beta}$, hence $\sigma^{1-M}\beta^{(M-1)/2} \asymp 1$, so $k(\sigma, \beta) \ge c$ for large β. Thus

$$\int_{\mathcal{R}} k(\sigma, \beta)\, d\sigma\, d\beta \ge c \int_B^\infty \int_{c_1\sqrt{\beta}}^{c_2\sqrt{\beta}} d\sigma\, d\beta = c(c_2 - c_1) \int_B^\infty \sqrt{\beta}\, d\beta = \infty.$$

Hence $\int_{\sigma>0} \int_{\beta \ge 2} k(\sigma, \beta)\, d\sigma\, d\beta = \infty$, and since $J(\sigma, \beta) < \infty$ the full integral $\int \tilde{p}(\Delta, \sigma, \beta \mid L_{1:M})\, d\Delta\, d\sigma\, d\beta$ diverges.

Conclusion. Under $\tilde{\pi}_1(\Delta, \sigma, \beta) = \Theta(\sigma)\,\Theta(\beta - 2)$ the posterior is **improper**.

3.3 (d) Posterior sampling with prior 2

We sample from $p(\Delta, \sigma, \beta \mid L_{1:M}, L^*)$ under such improper prior the factor β^{-2} penalises large β (which correspond to almost Gaussian tails) and also fixes the impropriety of the flat prior: in the problematic joint tail $\sigma \sim \sqrt{\beta}$ the posterior kernel becomes integrable, so the posterior is proper under $\tilde{\pi}_2$.

We use random-walk Metropolis,Hastings via `metrop` (mcmc) with 180{,}000 iterations (10{,}000 burn-in), proposal scales $(0.02, 0.12, 1.2)$ and acceptance rate ≈ 0.23.

Table 3: Calibration posteriors.

Parameter	Mean	Median	2.5%	97.5%
Delta (litres)	0.300	0.301	0.175	0.425
Sigma (litres)	2.176	1.344	0.527	6.482
Beta	24.606	6.941	2.153	112.715

The joint posterior (Figure 2, left) shows a weak negative association: larger systematic bias Δ tends to pair with slightly smaller tail weight β. When the device bias moves the centre further from L^*, less extra tail mass is needed to explain residuals. The dependence is mild, so Δ and β are close to independent.

looking at the marginal distributions there is evidence for:

Systematic bias (Δ). Median ≈ 0.301 litres, 95% CI [0.175, 0.425]. The entire interval lies above zero ($P(\Delta > 0) \approx 1$), giving strong evidence the device over-reports lung capacity by about 0.301 litres ($\sim 8.7\%$ of $L^* = 3.44$). This would inflate regression intercepts if left uncorrected.

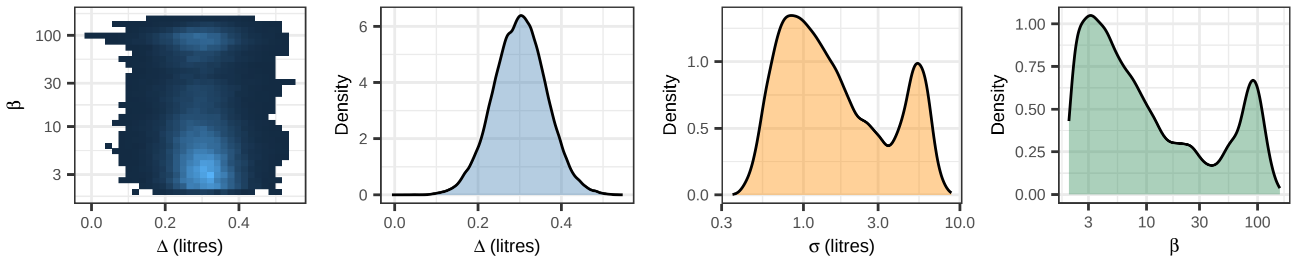

Figure 2: Calibration posteriors: joint (Δ, β) and marginals.

Measurement scale (σ). Median ≈ 1.34 litres, 95% CI [0.53, 6.48]. This is about 4.5 times larger than the $\sigma = 0.3$ assumed in Question 4, so the earlier regression standard errors and confidence intervals are too small by roughly this factor.

Tail weight (β). Median ≈ 6.9, 95% CI [2.2, 113], indicating considerable uncertainty about tail thickness. We find $P(\beta < 4) \approx 0.3$, and $P(\beta > 30) \approx 0.24$, leaving room for near-Gaussian behaviour. Most mass lies around $\beta \in [3, 10]$ (approximately Student-t with 5,19 df), consistent with occasional but not extreme outliers. The smallest and largest readings (2.48, 4.6) correspond to z-scores -0.94 and 0.64 respectively; a t_5 model fits such deviations roughly three times better than a Gaussian, pointing to mild outlier behaviour rather than very heavy tails.

4 Question 4 - lung capacity regression

4.1 (a) Normal-error model via glm

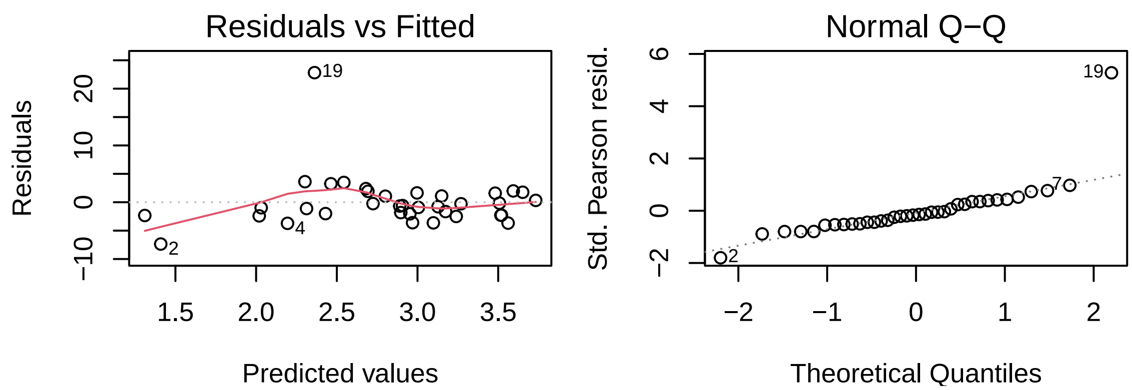

Figure 3: GLM diagnostics: residuals vs fitted and normal Q–Q plot.

Table 4: Normal-approx quantiles assuming known measurement SD = 0.3 L (dispersion fixed).

Term	Mean	SE	p	2.5%	25%	50%	75%	97.5%
(Intercept)	-2.905	0.879	0.001	-4.628	-3.498	-2.905	-2.312	-1.182
log_age	1.465	0.187	0.000	1.098	1.339	1.465	1.591	1.831
log_weight	0.058	0.183	0.751	-0.301	-0.065	0.058	0.182	0.417
cigarettes	-0.053	0.006	0.000	-0.065	-0.057	-0.053	-0.049	-0.041

I fit the Gaussian GLM $L_i = \beta_0 + \beta_1 \log(A_i) + \beta_2 \log(W_i) + \beta_3 \eta_i + \varepsilon_i$ using `glm()` with identity link and, as required, I compute standard errors and tail probabilities under the *known* measurement SD $\sigma = 0.3$ L (i.e. fixing the dispersion to σ^2).

From the standard `glm()` diagnostics, residuals are broadly centred at zero but the loess smooth is not flat (negative residuals at the smallest fitted values, slightly positive in the mid-range), suggesting mild mean mis-specification (weak nonlinearity or a missing interaction). The residual spread increases slightly with fitted capacity, indicating mild heteroskedasticity. The Q,Q plot shows a pronounced upper-tail departure driven by case 19 (very large positive residual), consistent with heavier-than-Gaussian positive errors; this point may be influential under squared-error loss, so Gaussian-model uncertainty may be optimistic.

In the coefficient table, the reported tail probabilities are *model-based* two-sided tests of $H_0 : \beta_j = 0$: $p_j = P(|Z_j| \geq |z_j^{obs}| \mid H_0,\ \varepsilon_i \sim N(0, 0.3^2))$ (using Wald z under fixed σ). They are **not** $P(\beta_j = 0 \mid \text{data})$ and they are valid only insofar as the Gaussian model (with fixed σ) is a reasonable approximation. Under this assumption, there is strong evidence that $\log(A)$ influences capacity positively and that η (cigarettes) influences capacity negatively (both $p \ll 0.05$), whereas there is no evidence that $\log(W)$ affects capacity ($p = 0.751$, with the Normal-approx interval spanning zero). Given the outlier and upper-tail non-normality, these significance levels should be interpreted cautiously, motivating the robust heavy-tailed analysis in part (b).

4.2 (b) Heavy-tailed regression via Stan

Table 5: Stan posterior summary for heavy-tailed regression (beta0..beta3).

Parameter	Mean	SE_mean	SD	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
beta0	2.057	0.017	0.807	0.314	1.555	2.115	2.607	3.518	2241.557	1.000
beta1	0.875	0.010	0.377	0.221	0.586	0.847	1.156	1.613	1475.352	1.002
beta2	-0.576	0.007	0.264	-1.096	-0.766	-0.559	-0.379	-0.113	1588.238	1.003
beta3	-0.097	0.000	0.008	-0.112	-0.103	-0.097	-0.091	-0.082	1818.187	1.002

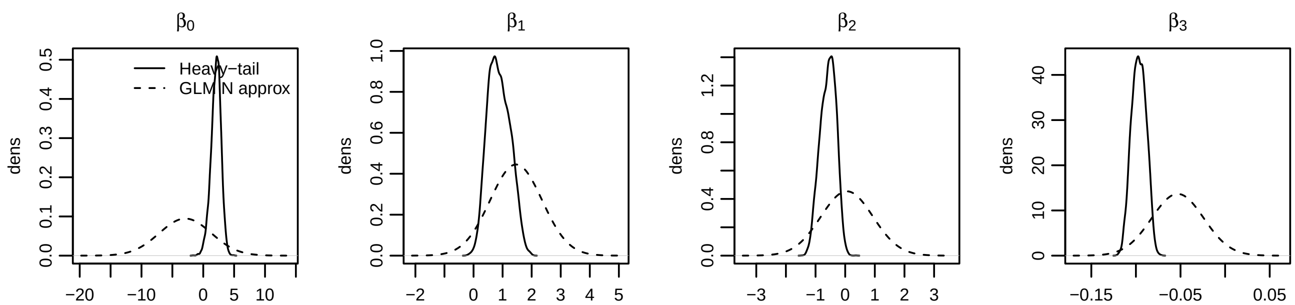

Figure 4: Marginal posteriors for $(\beta_0, \beta_1, \beta_2, \beta_3)$ under heavy-tailed errors ($\Delta = 0$, $\sigma = 0.3$, $\beta = 3$; solid), compared with the Normal approximation implied by the Gaussian GLM in part (a) (dashed).

This robust analysis uses the heavy,tailed error model from Question 3 with $\Delta = 0$, $\sigma = 0.3$ and $\beta = 3$, so each observation contributes $-\beta \log\{1 + ((L_i - \mu_i)/\sigma)^2\} - \log \sigma$ (up to a constant not depending on β_0:3). Unlike the Gaussian case, the implied loss has **bounded influence**: the score is proportional to $\frac{L_i - \mu_i}{\sigma^2 + (L_i - \mu_i)^2}$, so extreme residuals are downweighted (influence decays $\sim 1/(L_i - \mu_i)$ rather than growing linearly). This likelihood is not a standard exponential-family GLM, so `glm()` (IRLS for canonical families) cannot be used; we therefore implement it directly and sample the posterior in Stan with effectively flat priors on β_0:3 (approximated by very wide Normal priors).

MCMC diagnostics indicate reliable inference ($\hat{R} \approx 1$ and large n_{eff}; traceplots well mixed), so posterior summaries are stable Monte Carlo estimates of $p(\beta_j \mid \text{data})$. The marginals show strong evidence that $\beta_1 > 0$ and $\beta_3 < 0$ (posterior mass essentially entirely on one side of zero), implying a positive association of log(age) and a negative association of cigarette consumption with lung capacity, conditional on other covariates. In contrast, β_2 has substantial mass on both sides of zero, so there is no clear evidence of an additional log(weight) effect.

Compared with part (a), posterior centres are similar but uncertainty differs because the heavy-tailed likelihood reweights observations: the extreme upper-tail case seen in the Q,Q plot has much less leverage than under squared-error loss. With small N, Gaussian inference can be overly sensitive to such influential residuals and to tail misspecification; given the observed upper-tail non-normality, the heavy-tailed analysis is therefore more credible for interval estimation and for identifying effects that are robust rather than driven by one or two high-leverage points.

4.3 (c) Joint model for capacity and cancer

A coherent extension is to specify a *single* generative model for the mixed outcomes (L_i^{obs}, C_i) by introducing latent *true* lung capacity L_i^{true} and linking the two components through shared structure and full uncertainty propagation. Let $X_i = (\log A_i, \log W_i, \eta_i)$.

Measurement model (from Q3).
Model the device reading as heavy–tailed around the biased true value: $p(L_i^{obs} \mid L_i^{true}, \Delta, \sigma, \beta) \propto \left[1 + \frac{\{L_i^{obs} - (L_i^{true} + \Delta)\}^2}{\sigma^2}\right]^{-\beta}$, equivalently Student-$t$-like with $\nu = 2\beta - 1$ degrees of freedom (after rescaling). This ensures robustness to occasional extreme readings and keeps (Δ, σ, β) interpretable via the calibration experiment.

Latent capacity model.

$$L_i^{true} = \beta_0 + \beta^\top X_i + b_i + \varepsilon_i, \qquad b_i \sim N(0, \tau^2),$$

with ε_i Gaussian (or t if residual diagnostics suggest additional heavy tails beyond measurement error). The random effect b_i captures unobserved "frailty" affecting capacity.

Cancer model conditional on true capacity.

$$C_i \mid L_i^{true}, X_i, b_i \sim \text{Bernoulli}(\pi_i), \qquad \text{logit}(\pi_i) = \gamma_0 + \gamma^\top X_i + \gamma_4 L_i^{true} + \alpha b_i.$$

Including L_i^{true} allows cancer risk to depend on (latent) physiological capacity, while αb_i permits residual dependence between (L, C) from unmeasured factors not forced to act only through L_i^{true}.

The joint likelihood integrates out latent states:

$$p(L_i^{obs}, C_i \mid X_i) = \iint p(L_i^{obs} \mid L_i^{true}, \Delta, \sigma, \beta)\, p(L_i^{true} \mid X_i, b_i, \beta)\, p(C_i \mid L_i^{true}, X_i, b_i, \gamma)\, p(b_i \mid \tau)\, dL_i^{true}\, db_i,$$

so posterior predictive inference can target both margins and joint quantities (e.g. $P(C = 1,\ L^{obs} < \ell \mid X)$) with uncertainty propagated from measurement, regression, and latent heterogeneity.

Why this matters. Using L_i^{obs} directly in the cancer regression typically induces measurement-error bias in γ_4 (and for logistic links it is not a simple constant attenuation). Treating L_i^{true} as latent corrects this by integrating over $p(L_i^{true} \mid L_i^{obs}, \cdot)$.

Implementation (Stan). Sample $(\beta, \gamma, \tau, \alpha)$ and latent $(L_{1:N}^{true}, b_{1:N})$. Either fix (Δ, σ, β) at calibrated values, or propagate calibration uncertainty by placing informative priors for (Δ, σ, β) from the calibration posterior. Validate with posterior predictive checks for the tails of L^{obs} and calibration/discrimination for C, and compare to simpler models using LOO/WAIC.

5 Code appendix

```
knitr::opts_chunk$set(
  collapse = TRUE,
  comment = "#>",
  warning = FALSE,
  message = FALSE
)
include_solutions <- TRUE
options(lifecycle_verbosity = "quiet")
library(dplyr)
library(tidyr)
library(ggplot2)
library(kableExtra)
library(mcmc)
library(rstan)
library(gridExtra)

set.seed(20251203)
theme_set(theme_bw())

smoking <- read.csv("smoking.csv") |>
  mutate(
    smoker = cigarettes > 0,
    log_age = log(age),
    log_weight = log(weight)
  )

calibration <- read.csv("calibration.csv", header = FALSE)[[1]]
L_star <- 3.44
smoking$smoker <- smoking$cigarettes > 0
Ns <- sum(smoking$smoker)
Nns <- nrow(smoking) - Ns
Ns_c <- sum(smoking$smoker & smoking$cancer)
Nns_c <- sum(!smoking$smoker & smoking$cancer)

ap <- Ns_c + 1
bp <- Ns - Ns_c + 1
aq <- Nns_c + 1
bq <- Nns - Nns_c + 1

# Aliases for text references
alpha_p <- ap
beta_p <- bp
alpha_q <- aq
beta_q <- bq

# Posterior probability P(p > q | data)
prob_p_gt_q <- integrate(
  function(x) dbeta(x, ap, bp) * pbeta(x, aq, bq),
  lower = 0, upper = 1
)$value

# Posterior probability P(q > p | data)
prob_q_gt_p <- integrate(
  function(x) dbeta(x, aq, bq) * pbeta(x, ap, bp),
  lower = 0, upper = 1
```

```
)$value

Nc <- Ns_c + Nns_c
Ntot <- Ns + Nns

Nc <- Ns_c + Nns_c
Ntot <- Ns + Nns

# Marginal likelihood for M1: p > q
m1 <- 2 * integrate(
  function(p) {
    sapply(p, function(p_val) {
      integrate(
        function(q) {
          p_val^Ns_c * (1-p_val)^(Ns-Ns_c) * q^Nns_c * (1-q)^(Nns-Nns_c)
        },
        lower = 0, upper = p_val
      )$value
    })
  },
  lower = 0, upper = 1
)$value

# Marginal likelihood for M2: p = q
m2 <- beta(Nc + 1, (Ntot - Nc) + 1)

# Marginal likelihood for M3: p < q
m3 <- 2 * integrate(
  function(q) {
    sapply(q, function(q_val) {
      integrate(
        function(p) {
          p^Ns_c * (1-p)^(Ns-Ns_c) * q_val^Nns_c * (1-q_val)^(Nns-Nns_c)
        },
        lower = 0, upper = q_val
      )$value
    })
  },
  lower = 0, upper = 1
)$value

pq_grid <- tibble(
  x = rep(seq(0, 1, length.out = 400), 2),
  density = c(dbeta(seq(0, 1, length.out = 400), ap, bp),
              dbeta(seq(0, 1, length.out = 400), aq, bq)),
  parameter = rep(c("p = P(C|S)", "q = P(C|not S)"), each = 400)
)

counts_tbl <- tibble(
  group = c("Smokers", "Non-smokers"),
  N = c(Ns, Nns),
  cancer = c(Ns_c, Nns_c),
  no_cancer = N - cancer
)

posterior_summary <- tibble(
  parameter = c("p = P(C|S)", "q = P(C|not S)"),
```

```
  mean = c(ap / (ap + bp), aq / (aq + bq)),
  median = c(qbeta(0.5, ap, bp), qbeta(0.5, aq, bq)),
  lower95 = c(qbeta(0.025, ap, bp), qbeta(0.025, aq, bq)),
  upper95 = c(qbeta(0.975, ap, bp), qbeta(0.975, aq, bq))
)

evidence_tbl <- tibble(
  model = c("M1: p > q", "M2: p = q", "M3: p < q"),
  log_marginal = log(c(m1, m2, m3)),
  BF_vs_M2 = c(m1 / m2, 1, m3 / m2)
) |>
  mutate(BF_vs_best = exp(log_marginal - max(log_marginal)))
library(gridExtra)
library(grid)
library(ggplot2)

# Build the posterior plot (ggplot object)
pq_plot <- ggplot(pq_grid, aes(x = x, y = density, colour = parameter)) +
  geom_line(size = 0.8) +
  scale_colour_brewer(palette = "Dark2") +
  labs(x = "Probability", y = "Density", colour = NULL) +
  theme(legend.position = "top", text = element_text(size = 9))

# Build table grobs for stacking
table_theme <- ttheme_minimal(
  base_size = 9,
  core = list(fg_params = list(hjust = 0.5, x = 0.5),
              bg_params = list(fill = c("white", "grey95"), col = "black", lwd = 0.5)),
  colhead = list(fg_params = list(hjust = 0.5, x = 0.5),
                 bg_params = list(fill = "grey90", col = "black", lwd = 0.5))
)

t1 <- tableGrob(counts_tbl %>% mutate(across(where(is.numeric), as.character)), rows = NULL, theme = tabl
t2 <- tableGrob(posterior_summary %>% mutate(across(where(is.numeric), ~round(.x, 3))) %>% mutate(across(

# Left column: stacked tables
left_stack <- arrangeGrob(t1, t2, ncol = 1, heights = c(1, 1))

# Combine left (tables) and right (plot) into one figure
full <- arrangeGrob(left_stack, pq_plot, ncol = 2, widths = c(0.5, 0.5))

grid.draw(full)

# 1. Binomial factor (for the count likelihood)
binom_factor <- choose(Ns, Ns_c) * choose(Nns, Nns_c)

# 2. Likelihood kernel function (without binomial coeffs)
lik_kernel <- function(p, q) {
  (p^Ns_c * (1 - p)^(Ns - Ns_c)) *
    (q^Nns_c * (1 - q)^(Nns - Nns_c))
}

# 3. Calculate M1 (p > q)
integrand_m1 <- function(q, p) { 2 * lik_kernel(p, q) }

# Inner integral over q (0 to p), outer over p (0 to 1)
inner_int_m1 <- Vectorize(function(p) {
```

```r
  integrate(function(q) integrand_m1(q, p), lower = 0, upper = p)$value
})
m1_integral <- integrate(inner_int_m1, lower = 0, upper = 1)$value
m1 <- binom_factor * m1_integral

# 4. Calculate M2 (p = q) - Analytic solution
Nc      <- Ns_c + Nns_c
N_total <- Ns + Nns
m2      <- binom_factor * beta(Nc + 1, N_total - Nc + 1)

# 5. Calculate M3 (p < q)
# Same kernel, but integrate over the region 0 < p < q < 1
integrand_m3 <- function(q, p) { 2 * lik_kernel(p, q) }

inner_int_m3 <- Vectorize(function(p) {
  integrate(function(q) integrand_m3(q, p), lower = p, upper = 1)$value
})
m3_integral <- integrate(inner_int_m3, lower = 0, upper = 1)$value
m3 <- binom_factor * m3_integral

# Bayes factors (later used in text)
bf12 <- m1 / m2
bf13 <- m1 / m3
library(dplyr)
library(knitr)

## Marginal likelihoods
ml_vals <- c(m1, m2, m3)
ml_fmt  <- formatC(ml_vals, format = "e", digits = 2)
models  <- c("M1: p > q", "M2: p = q", "M3: p < q")

ml_tbl <- tibble(
  Model = models,
  `Marginal Likelihood` = ml_fmt
)

## Pairwise Bayes factors (i over j)
bf_mat <- matrix(c(
  1,        m1 / m2, m1 / m3,
  m2 / m1,  1,       m2 / m3,
  m3 / m1,  m3 / m2, 1
), nrow = 3, byrow = TRUE,
dimnames = list(models, models))

bf_df <- as_tibble(bf_mat) |>
  mutate(Model = models) |>
  select(Model, everything())

tbl1 <- kable(
  ml_tbl,
  format   = "latex",
  booktabs = TRUE
)

tbl2 <- kable(
  bf_df |> mutate(across(-Model, ~ round(.x, 2))),
  format   = "latex",
```

```r
  booktabs = TRUE
)

cat("
\\begin{minipage}{0.48\\linewidth}
\\centering
", tbl1, "
\\captionof{table}{Marginal likelihoods (scientific notation)}
\\end{minipage}\\hfill
\\begin{minipage}{0.48\\linewidth}
\\centering
", tbl2, "
\\captionof{table}{Pairwise Bayes factors (rows = numerator, columns = denominator)}
\\end{minipage}
")
log_post <- function(theta, data) {
  delta <- theta[1]
  sigma <- theta[2]
  beta <- theta[3]

  # Enforce constraints via prior
  if (sigma <= 0 || beta <= 2) {
    return(-Inf)
  }

  resid <- data - (L_star + delta)
  ll <- sum(-beta * log1p((resid / sigma) ^ 2) -
              log(sigma) + lgamma(beta) - lgamma(beta - 0.5) - 0.5 * log(pi))
  lp <- ll - 2 * log(beta)

  if (!is.finite(lp)) return(-Inf)
  lp
}

set.seed(20251203)
mcmc_out <- metrop(log_post, initial = c(0.3, 1.5, 6),
                   nbatch = 180000, scale = c(0.02, 0.12, 1.2), data = calibration)
calib_draws <- as.data.frame(mcmc_out$batch) |>
  transmute(
    delta = V1,
    sigma = V2,
    beta = V3
  ) |>
  slice(10001:n())

calib_summary <- tibble(
  parameter = c("Delta (litres)", "Sigma (litres)", "Beta"),
  mean = c(mean(calib_draws$delta), mean(calib_draws$sigma), mean(calib_draws$beta)),
  median = c(median(calib_draws$delta), median(calib_draws$sigma), median(calib_draws$beta)),
  lower95 = c(quantile(calib_draws$delta, 0.025),
              quantile(calib_draws$sigma, 0.025),
              quantile(calib_draws$beta, 0.025)),
  upper95 = c(quantile(calib_draws$delta, 0.975),
              quantile(calib_draws$sigma, 0.975),
              quantile(calib_draws$beta, 0.975))
)
```

```r
prob_beta_lt4 <- mean(calib_draws$beta < 4)
prob_beta_gt30 <- mean(calib_draws$beta > 30)
prob_delta_positive <- mean(calib_draws$delta > 0)
calib_summary |>
  mutate(across(where(is.numeric), ~round(.x, 3))) |>
  kable(col.names = c("Parameter", "Mean", "Median", "2.5%", "97.5%"),
        caption = "Calibration posteriors.",
        format = "latex",
        booktabs = TRUE) |>
  kable_styling(latex_options = "hold_position")
p_joint <- ggplot(calib_draws, aes(x = delta, y = beta)) +
  geom_bin2d(bins = 30) + scale_y_log10() +
  labs(x = expression(Delta~"(litres)"), y = expression(beta)) +
  theme(legend.position = "none", text = element_text(size = 7),
        axis.text = element_text(size = 6))

p_delta <- ggplot(calib_draws, aes(x = delta)) +
  geom_density(fill = "steelblue", alpha = 0.4) +
  labs(x = expression(Delta~"(litres)"), y = "Density") +
  theme(text = element_text(size = 7), axis.text = element_text(size = 6))

p_sigma <- ggplot(calib_draws, aes(x = sigma)) +
  geom_density(fill = "darkorange", alpha = 0.4) + scale_x_log10() +
  labs(x = expression(sigma~"(litres)"), y = "Density") +
  theme(text = element_text(size = 7), axis.text = element_text(size = 6))

p_beta <- ggplot(calib_draws, aes(x = beta)) +
  geom_density(fill = "seagreen", alpha = 0.4) + scale_x_log10() +
  labs(x = expression(beta), y = "Density") +
  theme(text = element_text(size = 7), axis.text = element_text(size = 6))

grid.arrange(p_joint, p_delta, p_sigma, p_beta, nrow = 1)
library(dplyr)

smoking <- read.csv("smoking.csv") |>
  mutate(
    log_age    = log(age),
    log_weight = log(weight)
  )

# Gaussian GLM with identity link.
# We use inverse-variance weights corresponding to sd = 0.3 L.
glm_fit <- glm(
  capacity ~ log_age + log_weight + cigarettes,
  data    = smoking,
  family  = gaussian(),
  weights = rep(1 / 0.3^2, nrow(smoking))
)

glm_coef <- coef(summary(glm_fit))

glm_table <- data.frame(
  term     = c("(Intercept)", "log(age)", "log(weight)", "cigarettes"),
  estimate = glm_coef[, "Estimate"],
  se       = glm_coef[, "Std. Error"],
  t_value  = glm_coef[, "t value"],
  p_value  = glm_coef[, "Pr(>|t|)"]
```

```r
) |>
  mutate(
    ci_low  = estimate - 1.96 * se,
    ci_high = estimate + 1.96 * se
  )

glm_display <- glm_table |>
  mutate(across(estimate:ci_high, ~ round(., 3))) |>
  as.data.frame()
par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 1), cex = 0.8)
plot(glm_fit, which = 1)  # residuals vs fitted: linearity + constant variance
plot(glm_fit, which = 2)  # normal Q,Q: normality of Pearson residuals
par(mfrow = c(1, 1))
#  1) Model-based inference with KNOWN sigma = 0.3
sig <- 0.3

# Fit (coefficients do not depend on sigma for Gaussian identity)
glm_fit <- glm(
  capacity ~ log_age + log_weight + cigarettes,
  data   = smoking,
  family = gaussian()
)

# Force known measurement variance in the SEs
s_fixed <- summary(glm_fit, dispersion = sig^2)
tab_fix <- coef(s_fixed)
params <- rownames(tab_fix)

# Under known sigma: use Wald z with Normal tails
z_fix <- tab_fix[, "Estimate"] / tab_fix[, "Std. Error"]
p_fix <- 2 * pnorm(-abs(z_fix))

# Normal-approx quantiles for each coefficient
norm_q <- function(mu, se, probs = c(0.025, 0.25, 0.5, 0.75, 0.975)) {
  as.numeric(qnorm(probs, mean = mu, sd = se))
}
q_fix <- t(mapply(norm_q, mu = tab_fix[, "Estimate"], se = tab_fix[, "Std. Error"]))
colnames(q_fix) <- c("2.5%", "25%", "50%", "75%", "97.5%")

tab_model <- data.frame(
  Term = params,
  Mean = tab_fix[, "Estimate"],
  SE   = tab_fix[, "Std. Error"],
  p    = p_fix,
  `2.5%`  = q_fix[, "2.5%"],
  `25%`   = q_fix[, "25%"],
  `50%`   = q_fix[, "50%"],
  `75%`   = q_fix[, "75%"],
  `97.5%` = q_fix[, "97.5%"],
  row.names = NULL,
  check.names = FALSE
) |>
  dplyr::mutate(dplyr::across(where(is.numeric), ~ round(.x, 3)))

#  print two compact tables
print_block <- function(df, caption_txt) {
  knitr::kable(
```

```r
    df,
    format   = "latex",
    booktabs = TRUE,
    caption  = caption_txt,
    align    = "lrrrrrrrr"
  ) |>
    kableExtra::kable_styling(latex_options = "hold_position", font_size = 8)
}

print_block(tab_model, "Normal-approx quantiles assuming known measurement SD = 0.3 L (dispersion fixed).

# Map GLM estimates into a vector with names matching the Stan parameters
beta_hat_glm <- c(
  beta0 = glm_coef["(Intercept)", "Estimate"],
  beta1 = glm_coef["log_age", "Estimate"],
  beta2 = glm_coef["log_weight", "Estimate"],
  beta3 = glm_coef["cigarettes", "Estimate"]
)

beta_se_glm <- c(
  beta0 = glm_coef["(Intercept)", "Std. Error"],
  beta1 = glm_coef["log_age", "Std. Error"],
  beta2 = glm_coef["log_weight", "Std. Error"],
  beta3 = glm_coef["cigarettes", "Std. Error"]
)

# Stan code for the heavy-tailed regression (log-density up to additive constant)
stan_code <- "
data {
  int<lower=1> N;
  vector[N] L_obs;
  vector[N] log_age;
  vector[N] log_weight;
  vector[N] cigs;
  real<lower=0> sigma_eps;   // fixed measurement scale
  real<lower=0> beta_tail;   // fixed tail parameter beta
}
parameters {
  real beta0;
  real beta1;
  real beta2;
  real beta3;
}
model {
  vector[N] mu;
  vector[N] u;

  // Very weakly informative priors approximating flat priors
  beta0 ~ normal(0, 100);
  beta1 ~ normal(0, 100);
  beta2 ~ normal(0, 100);
  beta3 ~ normal(0, 100);

  // Linear predictor
  mu = beta0 + beta1 * log_age + beta2 * log_weight + beta3 * cigs;

  // Heavy-tailed likelihood (up to a constant)
```

```
  u = (L_obs - mu) / sigma_eps;
  for (n in 1:N)
    target += - beta_tail * log1p(square(u[n])) - log(sigma_eps);
}
"

writeLines(stan_code, con = "q4b_heavytail_regression.stan")

rstan::rstan_options(auto_write = TRUE)
options(mc.cores = max(1L, parallel::detectCores() - 1L))

# Build the data list from the `smoking` dataframe used earlier
stan_data <- list(
  N = nrow(smoking),
  L_obs = smoking$capacity,
  log_age = smoking$log_age,
  log_weight = smoking$log_weight,
  cigs = smoking$cigarettes,
  sigma_eps = 0.3,
  beta_tail = 3
)

# Compile and sample
sm_model <- rstan::stan_model(file = "q4b_heavytail_regression.stan")

fit_heavy <- rstan::sampling(
  sm_model,
  data = stan_data,
  chains = 4,
  iter = 4000,
  warmup = 2000,
  seed = 123,
  refresh = 0
)

# Stan summary table without duplicated "parameter"

fit_sum_raw <- rstan::summary(
  fit_heavy,
  pars = c("beta0", "beta1", "beta2", "beta3")
)$summary

fit_summary_disp <- as.data.frame(fit_sum_raw)
fit_summary_disp$Parameter <- rownames(fit_sum_raw)
rownames(fit_summary_disp) <- NULL

fit_summary_disp <- dplyr::select(
  fit_summary_disp,
  Parameter, mean, se_mean, sd, `2.5%`, `25%`, `50%`, `75%`, `97.5%`, n_eff, Rhat
) |>
  dplyr::mutate(dplyr::across(where(is.numeric), ~ round(.x, 3)))

knitr::kable(
  fit_summary_disp,
  format    = "latex",
  booktabs  = TRUE,
  longtable = TRUE,
```

```r
  row.names = FALSE,
  caption   = "Stan posterior summary for heavy-tailed regression (beta0..beta3).",
  col.names = c("Parameter","Mean","SE_mean","SD","2.5%","25%","50%","75%","97.5%","n_eff","Rhat")
) |>
  kableExtra::kable_styling(latex_options = c("hold_position","repeat_header"), font_size = 8)


# Extract posterior draws
posterior_draws <- as.data.frame(fit_heavy, pars = c("beta0","beta1","beta2","beta3"))

param_names  <- c("beta0","beta1","beta2","beta3")
nice_labels  <- c(expression(beta[0]), expression(beta[1]), expression(beta[2]), expression(beta[3]))

par(mfrow = c(1, 4), mar = c(3.2, 3.2, 2.0, 0.8), cex = 0.75, mgp = c(1.8, 0.6, 0))
for (j in seq_along(param_names)) {
  par_name <- param_names[j]
  draws_j  <- posterior_draws[[par_name]]
  dens_j   <- density(draws_j)

  mean_glm <- beta_hat_glm[par_name]
  se_glm   <- beta_se_glm[par_name]

  x_min <- min(dens_j$x, mean_glm - 4 * se_glm)
  x_max <- max(dens_j$x, mean_glm + 4 * se_glm)

  plot(dens_j,
       xlim = c(x_min, x_max),
       main = as.expression(nice_labels[j]),
       xlab = "",
       ylab = "dens",
       cex.main = 0.9, cex.axis = 0.8, cex.lab = 0.8)

  curve(dnorm(x, mean = mean_glm, sd = se_glm),
        from = x_min, to = x_max, lty = 2, add = TRUE)

  if (j == 1) {
    legend("topright",
           legend = c("Heavy-tail", "GLM N approx"),
           lty = c(1, 2), bty = "n", cex = 0.75)
  }
}
par(mfrow = c(1, 1))
```

6 References